The Shadow Price of Neutrality

TODO(Isaiah): author

Abstract—TODO(Isaiah), Abstract, 0.3 page (claim → evidence):

- Problem: a long/short pod under a platform's neutrality mandate at a fixed 3%/yr risk budget → Sec. II.
- Approach: duals of the convex program at a root-found λ^*; three integer sensitivity estimators against certified secant intervals; big-M against perspective formulations → Sec. III.
- Q1 finding: liquidity and sector bounds carry the cost; factor neutrality is nearly free for α_{mom} → 0.16, 0.014, 0.964; Figs. 1–2.
- Q2 finding: with integers, no continuous estimator recovers the secant; the fixed-integer dual is exact only when the support does not change → Table II, 6.7×10^{-12}, 0.056.
- Q3 finding: the perspective relaxation is tighter, but default scaling defeats it; rescaling the weights fixes it → 0.057, 0.015, 3.3×10^{-4}; Fig. 5.

I. Introduction

TODO(Isaiah), I, 0.8 page (claim → evidence):

- Motivation and significance: a multi-manager platform imposes neutrality to protect its other books; each constraint costs the pod expected alpha; the dual is the price of that constraint → Sec. II.
- Related work: fundamental law and transfer coefficient [1], [2]; constraint attribution [3]; factor alignment [4], [5]; realistic shorts [6]; perspective reformulations [7], [8]; integer sensitivity [9], [10]; sparse portfolios [11]; course texts [12], [13].
- Questions Q1–Q3 (DESIGN §2) → Sec. IV.
- Contribution: costs at equal risk via duality at λ^*, with the alignment closed form $\sqrt{1-\varphi}$ → A1–A4, Fig. 2.
- Contribution: certified secant intervals that test integer sensitivity estimators against solver error → A8, Table II, 3.6×10^{-4}.
- Contribution: a support-stability account of when the fixed-integer dual is exact → 6.7×10^{-12} (P1), 0.056 (P3).
- Contribution: weight rescaling as the fix for the perspective cones' absolute tolerance → A9, Table III, 8.0×10^{-9}.
- Project type: problem formulation, reusing published formulations in a new setting.

II. Problem Setting

A. Decision problem

A long/short equity pod at a multi-manager platform receives each month a forecast α_i of every stock's alpha, its expected return beyond what its factor exposures explain, and chooses weights w as fractions of assets under management (AUM, \$1B). The platform caps the pod's ex-ante volatility at 3%/yr and imposes a mandate that protects its other books. A portfolio's factor exposure is $y = B^\top w$, where B holds each stock's loadings on six factors; its beta is the exposure y_{mkt} to the market factor. Dollar neutrality bounds the net weight $|1^\top w|$, beta neutrality bounds $|y_{\text{mkt}}|$, style neutrality bounds the other five exposures, and sector neutrality bounds each sector's net weight $|S_j^\top w|$. Gross leverage $1^\top(w^+ + w^-)$ is the total long plus short weight. A stock's average daily volume (ADV) is the median daily dollar volume over the last 63 trading days, and the liquidity cap limits each position to the fraction θ of ADV: $u_i = \min(u, \theta\,\text{ADV}_i/\text{AUM})$. Shorts pay a borrow cost c per month and every trade a one-way trading cost κ; the pod starts from cash ($w_0 = 0$), so the trading cost is κ times gross. Table I lists the base mandate.

B. Data and risk model

Daily adjusted prices and volumes of the current S&P 500 constituents (500 tickers, 2005-01-04 to 2025-12-31, 5282 trading days) come from yfinance, GICS sectors from the constituent list, and the daily Fama–French five factors and momentum from the French data library [14]–[16]. At each of the 241 month-ends from 2005-12 to 2025-12, the risk model regresses each eligible stock's trailing 252 daily returns on the six factors. B holds the loadings, $D = \text{diag}(d)$ the monthly residual variances, floored at the 5th cross-sectional percentile, and F the monthly factor covariance, so $\Sigma = BFB^\top + D \succ 0$. The eligible universe has 373–499 names. The continuous experiments use the dates D12 = {2008-10, 2009-03, 2011-08, 2013-06, 2015-08, 2017-12, 2018-12, 2020-03, 2020-11, 2022-06, 2023-03, 2025-04}, chosen to span calm and stressed markets.

C. Alphas and alignment

The momentum alpha α_{mom} is each stock's return over the year before the date, skipping the last month, z-scored across the universe, winsorized at ±3, re-standardized, and scaled as $\alpha_i = 0.03\sqrt{d_i}\, z_i$. It is an input signal, not a strategy; no backtest is run. A D^{-1}-weighted least-squares regression of α on the loadings splits it into a residual $\alpha_\perp = \alpha - B(B^\top D^{-1} B)^{-1} B^\top D^{-1}\alpha$ and a factor-spanned part $\alpha_\parallel = \alpha - \alpha_\perp$. The information ratio $\sqrt{\alpha^\top \Sigma^{-1}\alpha}$ is the best attainable ratio of expected alpha to ex-ante risk. The alignment share $\varphi = \alpha_\parallel^\top \Sigma^{-1}\alpha_\parallel / \alpha^\top \Sigma^{-1}\alpha \in [0,1]$ is the fraction of its square carried by the factors. For comparisons, $\alpha_\perp$ and $\alpha_\parallel$ are rescaled to α_{mom}'s information ratio. The transfer coefficient of a portfolio is the correlation, in

the Σ metric, between w and the unconstrained direction $\Sigma^{-1}\alpha$: $\mathrm{TC} = \alpha^\top w/(\sqrt{\alpha^\top\Sigma^{-1}\alpha}\sqrt{w^\top\Sigma w})$ [2].

D. The convex problem P-RB

With $w = w^+ - w^-$, $w^\pm \geq 0$, and σ_m the monthly volatility budget,

$$\begin{aligned}\max\ & \alpha^\top w - c\,1^\top w^- - \kappa\,1^\top(w^+ + w^-)\\ \text{s.t. }& y^\top F y + \textstyle\sum_i d_i w_i^2 \leq \sigma_m^2, \quad y = B^\top w,\\ & |1^\top w| \leq \varepsilon_{\mathrm{net}},\ |y_{\mathrm{mkt}}| \leq \varepsilon_\beta,\ |y_k| \leq b_k,\ |S_j^\top w| \leq s_j,\\ & 1^\top(w^+ + w^-) \leq G, \quad w_i^+, w_i^- \leq u_i.\end{aligned} \tag{1}$$

Absolute values and costs are linear through the split $w^\pm$; because $c, \kappa > 0$, at most one of $w_i^\pm$ is positive at an optimum. The optimal value A^* is net expected alpha per month. Each bound b_j has a dual μ_j (the sum of its two one-sided multipliers) and an elasticity $e_j = \mu_j b_j / A^*$, the percent change in alpha per percent change in the bound; a constraint binds if $e_j \geq 10^{-4}$. The risk row's multiplier is written λ^*. It is the Lagrange multiplier of the risk budget: the price of risk at which a mean-variance investor chooses the same portfolio (A1). Fixing the risk budget rather than λ compares constraint costs at equal risk, so relaxing a constraint cannot buy alpha by levering up.

E. The integer problem P-MI

P-MI adds a cardinality limit K and a minimum position $\ell = 0.25\%$, moves the risk row into the objective at its multiplier, and raises the position cap to $u = 6\%$:

$$\begin{aligned}\max\ & \alpha^\top w - \lambda^*\big(y^\top F y + \textstyle\sum_i d_i w_i^2\big)\\ & \quad - c\,1^\top w^- - \kappa\,1^\top(w^+ + w^-)\\ \text{s.t. }& \text{the mandate of (1) without the risk row,}\\ & \ell z_i^\pm \leq w_i^\pm \leq u_i z_i^\pm,\ z_i^+ + z_i^- \leq 1,\ \textstyle\sum_i (z_i^+ + z_i^-) \leq K,\\ & z^\pm \in \{0,1\}^N.\end{aligned} \tag{2}$$

λ^* is P-RB's multiplier on the same universe at the base mandate with $u = 6\%$, held fixed for that date and universe. P-MI is then the same pod with integer constraints added: a Lagrangian relaxation of the risk row, with zero gap in the convex case. The big-M formulation keeps $\sum_i d_i w_i^2$ with $M = u_i$. The perspective formulation replaces it by $\sum_i d_i s_i$ with rotated cones $w_i^2 \leq s_i(z_i^+ + z_i^-)$, $s_i \geq 0$. The rescaled perspective is the same model with weights in units of the largest position bound and each cone written in its name's bound, $t_i = w_i/u_i$, $t_i^2 \leq s_i'\zeta_i$. Its feasible set and objective are unchanged; only the quantities Gurobi's absolute tolerances act on differ (A9).

F. Problem classes

P-RB is a convex quadratically constrained program (QCQP); its risk row is the second-order cone $\|(L_F^\top y, \sqrt{d}\circ w)\|_2 \leq \sigma_m$ with $F = L_F L_F^\top$. When the risk row binds the optimum is unique, because a linear objective over a strictly convex set attains its maximum at a single point.

TABLE I
Base mandate (P-RB) and the P-MI parameters. AUM: assets under management; ADV: average daily dollar volume.

Parameter	Base
Risk budget (ex-ante volatility)	3%/yr
Gross leverage G	2
Net (dollar) $\varepsilon_{\mathrm{net}}$	±2%
Beta ε_β	±0.05
Each style b_k	±0.10
Each sector s_j (net)	±2%
Position cap u (P-MI)	3% (6%)
Liquidity θ	5% of ADV
AUM	\$1B
Borrow c	30 bp/yr
Trading κ (one-way)	5 bp
Minimum position ℓ (P-MI)	0.25%

The big-M P-MI is a convex mixed-integer QP, every node relaxation a convex QP; the perspective P-MI is a mixed-integer second-order cone program.

III. Solution Approach and Analysis

A. Algorithms

P-RB is solved as the mean-variance QP $\max\ \alpha^\top w - \lambda w^\top\Sigma w$ − costs by Gurobi's dual simplex at feasibility and optimality tolerances 10^{-9}, with λ found by Brent's method on $\log\lambda$ so that ex-ante volatility equals the budget to relative 10^{-10}. A1 makes this exact, and dual simplex returns exact active sets and duals; if $\lambda = 10^{-6}$ still leaves volatility below the budget, the risk row is slack. Clarabel's primal–dual interior-point method [17] (tolerances 10^{-12}) solves the explicit cone form of (1) as a cross-check. It also computes the perspective relaxation of (2) ($z \in [0,1]$), whose central finite differences give its slope; Clarabel's raw duals are kept as diagnostics. The big-M relaxation and the fixed-integer QP (P-MI with z fixed at the incumbent) are QPs solved by dual simplex. Gurobi's branch and bound solves P-MI [18], with its MIQCP method left at the default for the perspective model. The P-MI objective is multiplied by $s_{\mathrm{obj}} = 10^4$ so that its value is of order one relative to the solver's tolerances, and every value is reported in original units. Effort is Gurobi's deterministic Work, with WorkLimit $W^* = 340$ ($2W^*$ in I1 and I2). The incumbent is polished: the fixed-integer QP at its z is re-solved at tolerances 10^{-9}, and that value is V.

B. Analysis

A1 (risk budget and mean-variance). With multiplier μ for the risk row and ν for the mandate rows, the Lagrangian of (1) is $\alpha^\top w$ − costs − $\mu(w^\top\Sigma w - \sigma_m^2)$ − $\nu^\top$(mandate). Its stationarity condition is that of the mean-variance problem with $\lambda = \mu$, and under Slater's condition (A5) the KKT conditions are necessary and sufficient. So the mean-variance optimum at $\lambda = \mu_{\mathrm{risk}}$ solves P-RB: the risk budget is an ε-constraint and λ^* its scalarization weight.

A2–A3 (equality case). With only $Aw = 0$ and a binding risk row, stationarity gives $w^* = \Sigma^{-1}g/(2\mu)$ with $g = \alpha - A^\top\nu$; $Aw^* = 0$ gives $\nu = (A\Sigma^{-1}A^\top)^{-1}A\Sigma^{-1}\alpha$; the binding row gives $\mu = \sqrt{g^\top\Sigma^{-1}g}/(2\sigma_m)$ and $A^* = \sigma_m\sqrt{g^\top\Sigma^{-1}g}$. Divided by the unconstrained value $\sigma_m\sqrt{\alpha^\top\Sigma^{-1}\alpha}$, the retained fraction $\sqrt{g^\top\Sigma^{-1}g/\alpha^\top\Sigma^{-1}\alpha}$ equals TC at w^*. It is at most one by Cauchy–Schwarz in the Σ^{-1} metric, since g is the Σ^{-1}-orthogonal projection of α onto $\{v : A\Sigma^{-1}v = 0\}$.

A4 (alignment). By Woodbury, $B^\top\Sigma^{-1} = F^{-1}(F^{-1} + B^\top D^{-1}B)^{-1}B^\top D^{-1}$. The regression residual satisfies $B^\top D^{-1}\alpha_\perp = 0$, hence $B^\top\Sigma^{-1}\alpha_\perp = 0$ and $\Sigma^{-1}\alpha_\perp = D^{-1}\alpha_\perp$. Under the risk budget alone, $\alpha_\perp$'s optimum therefore has zero factor exposure, and factor bounds have zero shadow price. Because $\alpha_\parallel = B\beta$, the cross term $\alpha_\perp^\top\Sigma^{-1}\alpha_\parallel$ vanishes and $\alpha^\top\Sigma^{-1}\alpha = \alpha_\perp^\top D^{-1}\alpha_\perp + \alpha_\parallel^\top\Sigma^{-1}\alpha_\parallel$. A2 with $A = B^\top$ gives $g = \alpha_\perp$, so full factor neutrality retains $\sqrt{1-\varphi}$ of the alpha.

A5 (envelope). $w = 0$ strictly satisfies the risk row and the other constraints are affine, so Slater's condition holds, and strong duality with it. $A^*(b)$ is the perturbation function of a convex program, hence concave, and nondecreasing because loosening b enlarges the feasible set [12]. $dA^*/db_j = \mu_j$ wherever μ_j is unique (for example under LICQ), including across nondegenerate active-set changes, where $\mu_j(b)$ has a kink but A^* does not.

A6 (perspective tightness). For $z \in (0, 1]$, $s \geq w^2/z \geq w^2$, so in the relaxation the perspective's risk penalty is at least big-M's and its value at most big-M's: its bound is at least as tight. The perspective of a convex function is convex, and the perspective reformulation is the closure of the perspective cuts [7], [8].

A7 (one-asset toy). For $\max aw - \lambda dw^2$ over $w \in \{0\} \cup [\ell, u]$, $w \leq b$, the value jumps by $a\ell - \lambda d\ell^2$ at $b = \ell$. Below ℓ, the big-M relaxation's slope is $a - 2\lambda db$, the perspective relaxation's (minimizing s over $z \in [w/u, w/\ell]$) is $a - \lambda d\ell$, and the fixed-integer dual is 0, which is locally correct. A tighter bound need not give a better slope.

A8 (integer value functions). The P-MI value $U^*(b)$ is the upper envelope, over support patterns, of concave pieces: nondecreasing by feasible-set inclusion, but in general neither concave nor continuous. The relaxation dual is the slope of an upper-bounding concave function, the fixed-integer dual the slope of the active piece, and a re-solve gives a secant across pieces. For a bound perturbed by $\pm\delta$, $V \leq U^* \leq B_{\text{eff}}$ holds at each point, with V the polished value, B the final bound and $B_{\text{eff}} = \max(B, V) + 10^{-6}|V|$ (the slack covers Gurobi's tolerances). This gives the certified interval

$$L = \frac{V_+ - B_{\text{eff},-}}{2\delta} \leq \frac{U_+^* - U_-^*}{2\delta} \leq \frac{B_{\text{eff},+} - V_-}{2\delta} = H, \quad (3)$$

with point slope $s = (V_+ - V_-)/(2\delta)$. A case is binding iff $L > 0$; $H < 0$ is impossible at true optima and flags solver error. In I1 and I2 both formulations are solved at every point, V is the larger polished value and B the smaller bound (combined certification).

A9 (scaling). An absolute tolerance τ on a quantity of size m is a relative error τ/m, so rescaling a row or the objective gives an equivalent problem with different effective tolerances. The variance row has size $\sigma_m^2 = 7.5\times10^{-5}$. The perspective cones $w_i^2 \leq s_i\zeta_i$ have s_i between ℓ^2 and u_i^2, small against Gurobi's absolute feasibility tolerance 10^{-6}. Rescaling the weights by each name's bound makes the cones' tolerance relative. The objective effect of cone violation, $\lambda^* \sum_i d_i \max(0, w_i^2/\zeta_i - s_i)$ at the raw incumbent, is reported relative to $|V|$.

C. Finite differences and noise

The perspective relaxation's slope is the central difference $(V(b + \delta_r) - V(b - \delta_r))/(2\delta_r)$ of Clarabel values at $\delta_r = 10^{-3}\, b$. The value is the objective at Clarabel's (w, ζ) with each s_i set to w_i^2/ζ_i, so every cone holds exactly. A relaxation fails if its residuals exceed 10^{-8} or its duality gap exceeds $10^{-9}\max(|V|, |V_{\text{ref}}|)$, where V_{ref} is the value at the unperturbed bound. That accuracy bounds the difference's noise by $N = 10^{-9}\max(|V|, |V_{\text{ref}}|)/\delta_r$. The three-point one-sided slopes are recorded too; if they differ by more than $\max(10^{-4}|\hat{s}|, 8N)$ the case is flagged as a kink.

D. Validation

The test suite (499 tests) and logged check runs confirm the analysis numerically, each check with a negative control. The root-found QP and the Clarabel QCQP agree on A^* to 1.2×10^{-10}, on λ^* to 3.8×10^{-8} and on w to 4.4×10^{-7} (A1). The equality cases match A2–A3 to 8.9×10^{-16}, and the A4 split holds to 2.1×10^{-16}. On 202 random one-asset toys (8279 grid points) both P-MI formulations match A7's value function to 1.4×10^{-17}. The 3 points just below ℓ, within Gurobi's feasibility tolerance, where the solver may hold a name at $b < \ell$, are excluded and counted. Certified intervals contain the exact slope on 4 toy and 4 small MIQP cases solved to zero gap, and the $H < 0$ flag fires on a constructed case. An independent feasibility checker, sharing no code with the optimizer, fails on constructed violations and passes every reported solution. P-MI with $K \geq N$ and $\ell = 0$ equals P-RB to 1.5×10^{-15}. The slope method was gated before I1 on 200 random toys, with 0 misses at 8399 points, 0 of 39 objective-scale pairs outside tolerance, and a failed fraction of 4.7×10^{-5}.

IV. Results

A. Q1: Constraint Costs in the Convex Problem

Setup. C1 solves P-RB at the base mandate ($u = 3\%$) on the full universe at the 12 D12 dates for each of the three alphas (36 solves). C2 sweeps one constraint family at a time with all others at base (2040 solves). The criterion is the elasticity e_j, read from exact duals, which are validated

as derivatives below. Statistics are over the D12 dates for α_{mom} ($n = 12$) unless stated.

Which constraints bind. At base, A^* has median 30.5 bp/month (range 22.0–38.3), and the pod keeps a median 0.65 (0.51–0.77) of the alpha of the unconstrained, costless optimum. The risk budget binds in all but 4 of the 36 C1 solves, all of them $\alpha_{\parallel}$. The liquidity cap binds at all 12 dates with median elasticity 0.16, sector bounds at 12 (median 0.014) and dollar neutrality at 10 (median 1.3×10^{-3}). Style bounds bind at 4 dates and the position cap at 2; beta and gross bind at none. Fig. 1 shows the cost curves. At the tightest grid points the pod keeps a median 0.744 of its alpha at $\theta = 2.5\%$ (range 0.52–0.95), 0.961 under exact sector neutrality (0.89–0.98), 0.956 with every style exposure at zero, 0.997 under exact dollar neutrality and 0.999 under exact beta neutrality.

What determines the cost. Factor bounds are cheap because α_{mom} is weakly aligned with the factors: its alignment share φ is 0.011–0.070 over D12. Under idealized full factor neutrality (risk budget only) the retained fraction equals $\sqrt{1-\varphi}$ (A4), at least 0.964 for α_{mom} (Fig. 2). The numerical and closed-form fractions agree to 7.5×10^{-8} over 36 solves, and A^* to 1.6×10^{-14} relative over 72. The two extremes bracket this: $\alpha_{\perp}$ has $\varphi \le 3.1 \times 10^{-33}$ and loses nothing, while $\alpha_{\parallel}$ has $\varphi = 1.0$ and loses everything. At the base mandate the points lie well below the curve, so the cost comes from the caps and the costs, not from neutrality. The attribution of Fig. 3 agrees: the position and liquidity caps carry a median share 0.44 of the attributed loss, trading costs 0.30, sector bounds 0.087, borrow 0.070, and names held at zero 0.051.

Duals as derivatives. For every binding constraint at every D12 date (115 checks), the central difference at $\delta = 1\%$ of the bound matches μ_j within 1%. The 64 stencils whose dual vanishes at an end, or whose forward and backward differences differ by more than 1% of the central one, are rechecked at $\delta = 0.1\%$. The largest checked error is 3.1×10^{-4}, and 0 exceed 1%. Of the flagged points, 35 have an active-set change within the stencil (a mandate row, a cap, or a name entering or leaving the support) and 29 are curvature flags. Shrinking the step tenfold shrinks $|F - B|$, the gap between the forward and backward differences, by a factor of 0.090–0.11 at all 64 flagged points, as on a smooth piece: at this resolution $A^*(b)$ is differentiable even across active-set changes, as A5 predicts. Along all 312 C2 sweeps, A^* never decreases as a bound loosens (1728 pairs) and the secant slopes never increase (1104 triples) beyond tolerance: 0 failures.

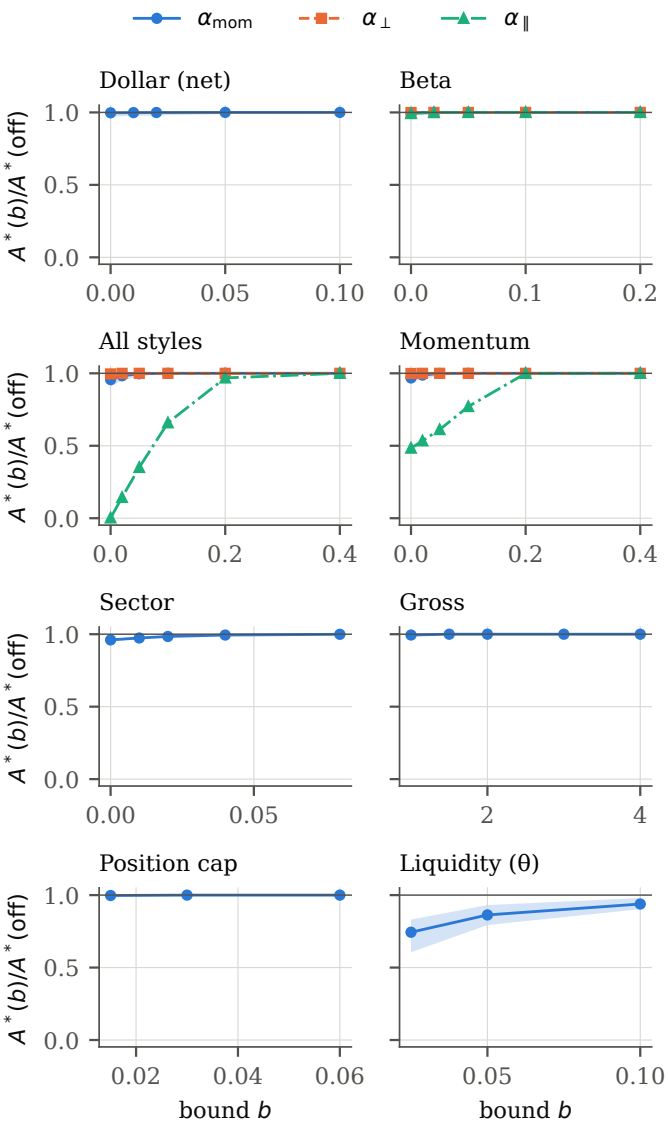


Fig. 1. Cost curves (C2): retained alpha $A^*(b)/A^*(\text{off})$ against each family's bound, all other constraints at base. Solid: the α_{mom} median with its interquartile band over the 12 D12 dates; for factor families, the $\alpha_{\perp}$ (dashed) and $\alpha_{\parallel}$ (dash-dotted) medians.

B. Q2: Sensitivity under Integrality

Setup. I1 solves P-MI at 2009-03, 2017-12 and 2020-03 on the top 250 names by ADV, with $K = \text{round}(fS)$ for $f \in \{0.25, 0.5, 0.75\}$ and S the continuous support size. Four constraints are perturbed by $\delta = 10\%$ of their bound: sector and dollar at 2%, all styles jointly at 0.10, and the momentum bound at 0.025; at 2017-12 also by $\delta = 5\%$. That gives 48 cases, 36 at $\delta = 10\%$ and 12 at $\delta = 5\%$. Each point is solved with both formulations at MIPGap 10^{-6} under combined certification. The estimators are the big-M relaxation dual, the perspective relaxation slope and the fixed-integer dual. The criterion is each estimator's relative error $|\hat{\mu} - s|/|s|$ over binding cases, and whether $\hat{\mu} \in [L, H]$.

Certification. No certified interval is violated (0 of 144 points), and the polished values of the two formulations were identical at all 144 I1 points; the bound came from big-M at 62 points and from the rescaled perspective at 82. At $\delta = 10\%$, 23 of 36 cases bind (sector 9, momentum 6, dollar 5, all styles 3). In the 13 non-binding cases at $\delta = 10\%$, and the 1 at $\delta = 5\%$, all three estimators are

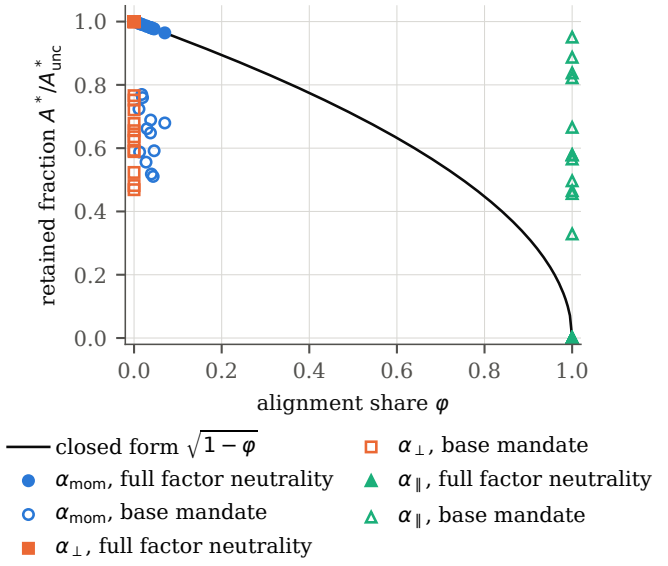


Fig. 2. Retained fraction A^*/A^*_{unc} against the alignment share φ (C1). Curve: the closed form $\sqrt{1-\varphi}$ (A4). Filled: idealized full factor neutrality with the risk budget only (three alphas × 12 dates). Hollow: the base mandate. Both are relative to the unconstrained, costless P-RB.

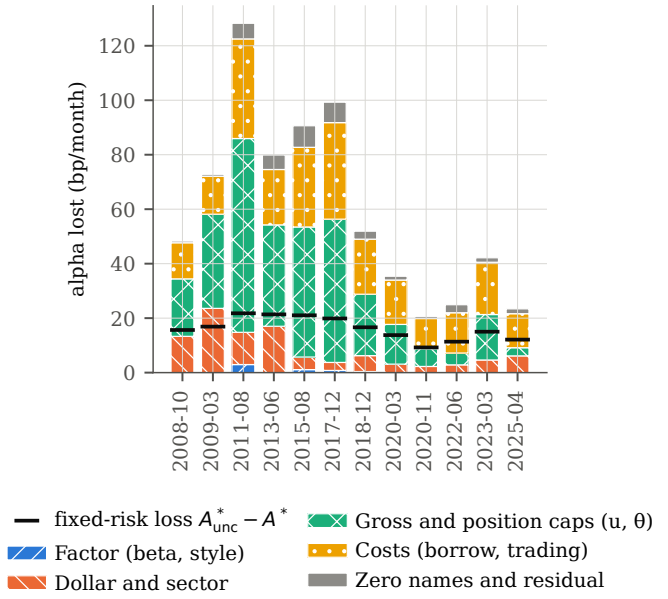


Fig. 3. Constraint attribution [3] for α_{mom} at the base mandate, evaluated at λ^*, by D12 date (dates with a slack risk budget omitted). The stacked total exceeds the fixed-risk loss $A^*_{\text{unc}} - A^*$ (markers) because the unconstrained portfolio at λ^* carries more than the 3%/yr budget.

zero. The intervals are narrow but not points: at $\delta = 10\%$ the relative width $(H - L)/s$ has median 3.6×10^{-4} (range 1.2×10^{-4}–0.020), set by B_{eff}'s slack. Lying in $[L, H]$ therefore means agreeing with the secant to within that width.

Estimator errors. Table II gives the errors at $\delta = 10\%$ ($n = 23$). The perspective slope has the smallest median error, 0.013 (interquartile range 6.8×10^{-3}–0.039, max 0.82). The fixed-integer dual follows with 0.023 (4.5×10^{-3}–0.067, max 0.82), and the big-M dual is worst, 0.073 (0.023–0.28, max 4.2). Only 4 estimates, all fixed-integer duals, lie in $[L, H]$. The convex baseline's dual is within a median 3.7×10^{-4} (max 4.2×10^{-3}, $n = 24$) of its own secant at $\delta = 10\%$, so integrality, not the step, accounts for the error. Only the big-M error falls as f, and so K/S, rises; the perspective error is flat from $f = 0.5$, and the fixed-integer error is not monotone. The errors depend on δ. At 2017-12 ($n = 11$ binding of 12 at each step), the median errors at $\delta = 5\%$ and 10% are 0.24 and 0.073 for big-M, 0.052 and 0.016 for the perspective, and 0.014 and 0.023 for the fixed-integer dual.

When the fixed-integer dual is exact. Each case is classed by comparing, at $b - \delta$, b and $b + \delta$, the support of V's incumbent (sign included) and the fixed-integer QP's active set. P1 means both unchanged, P2 the support unchanged but the active set changed, P3 the support changed. Of the 34 binding cases, 3, 2 and 18 are P1, P2 and P3 at $\delta = 10\%$, and 4, 1 and 6 at $\delta = 5\%$. In P1 the fixed-integer error is at most 6.7×10^{-12} ($n = 7$; predicted $\leq 10^{-6}$), and in P2 at most 2.9×10^{-4} ($n = 3$; predicted $\leq 4.2 \times 10^{-3}$, the baseline's largest secant error). In P3 its median is 0.056 and its max 0.82 ($n = 24$). Every fixed-integer estimate in $[L, H]$ (4 at $\delta = 10\%$, 5 at 5%) is in P1 or P2. The relaxation slopes fail differently. At $\delta = 10\%$ the perspective error is 0.011 in P1 ($n = 3$) and 0.014 in P3 ($n = 18$), and it rises with the relaxation's gap at b: the Spearman rank correlation (average ranks) between gap and error is 0.64 for the perspective and 0.50 for big-M ($n = 23$).

Value functions. I2 sweeps the sector bound, the I1 constraint with the largest median secant at $\delta = 10\%$ (1.364×10^{-3}, against 1.316×10^{-3} for the momentum bound), over 41 points on $[0, 2b]$ at 2017-12 ($K = 67$) and 2020-03 ($K = 79$), with 250 names (Fig. 4). The P-MI value lies below the continuous baseline, and its grid secants are not monotone. At 2017-12, 12 of 39 grid triples are certified nonconcave, $V(b_{k-1}) + V(b_{k+1}) > 2B_{\text{eff}}(b_k)$; at 2020-03, 9 of 39. Every nonconcavity seen in V alone is certified (12 and 9). The support changes between 16 of 40 and 17 of 40 neighbouring points, by 2–4 names. At $0.5b$, b and $1.5b$ ($n = 6$) the median errors are 0.32 (big-M), 0.078 (perspective) and 0.059 (fixed-integer), and none lies in its interval, whose relative width is at most 9.9×10^{-4}. The baseline's dual matches its envelope slope within 1% at all 80 grid points.

Slope accuracy. None of the 240 I1 perspective relaxations failed and 0 kinks were flagged; Clarabel's raw duals agree with the finite-difference slope to a median 7.2×10^{-9} (max 2.8×10^{-4}). The slopes do not depend on the objective scale: at 10 times s_{obj} they agree within tolerance in all 12 gated cases at 2009-03 and in all 36 remaining cases (binding cases: max relative difference 2.6×10^{-7}, $n = 25$).

TABLE II

I1 estimator errors $|\hat{\mu} - s|/|s|$ at $\delta = 10\%$ over binding cases, by f ($K = \text{round}(fS)$ differs by date), with the number in the certified interval $[L, H]$ and the median relative interval width of the row's cases. Bottom: the convex baseline's dual against its own secant at $\delta = 10\%$.

estimator	f	n	median err.	max err.	in $[L, H]$	median $(H - L)/s$
big-M relaxation dual	0.25	7	2.6×10^{-1}	4.2×10^{0}	0	6.2×10^{-4}
big-M relaxation dual	0.5	8	1.4×10^{-1}	1.2×10^{0}	0	3.2×10^{-4}
big-M relaxation dual	0.75	8	1.9×10^{-2}	3.0×10^{-1}	0	3.2×10^{-4}
perspective relaxation slope	0.25	7	4.5×10^{-2}	8.2×10^{-1}	0	6.2×10^{-4}
perspective relaxation slope	0.5	8	8.2×10^{-3}	1.0×10^{-1}	0	3.2×10^{-4}
perspective relaxation slope	0.75	8	8.4×10^{-3}	5.2×10^{-2}	0	3.2×10^{-4}
fixed-integer dual	0.25	7	4.8×10^{-2}	7.8×10^{-1}	1	6.2×10^{-4}
fixed-integer dual	0.5	8	6.0×10^{-2}	8.2×10^{-1}	0	3.2×10^{-4}
fixed-integer dual	0.75	8	5.2×10^{-3}	2.9×10^{-2}	3	3.2×10^{-4}
convex baseline dual vs. secant	all	24	3.7×10^{-4}	4.2×10^{-3}	–	–

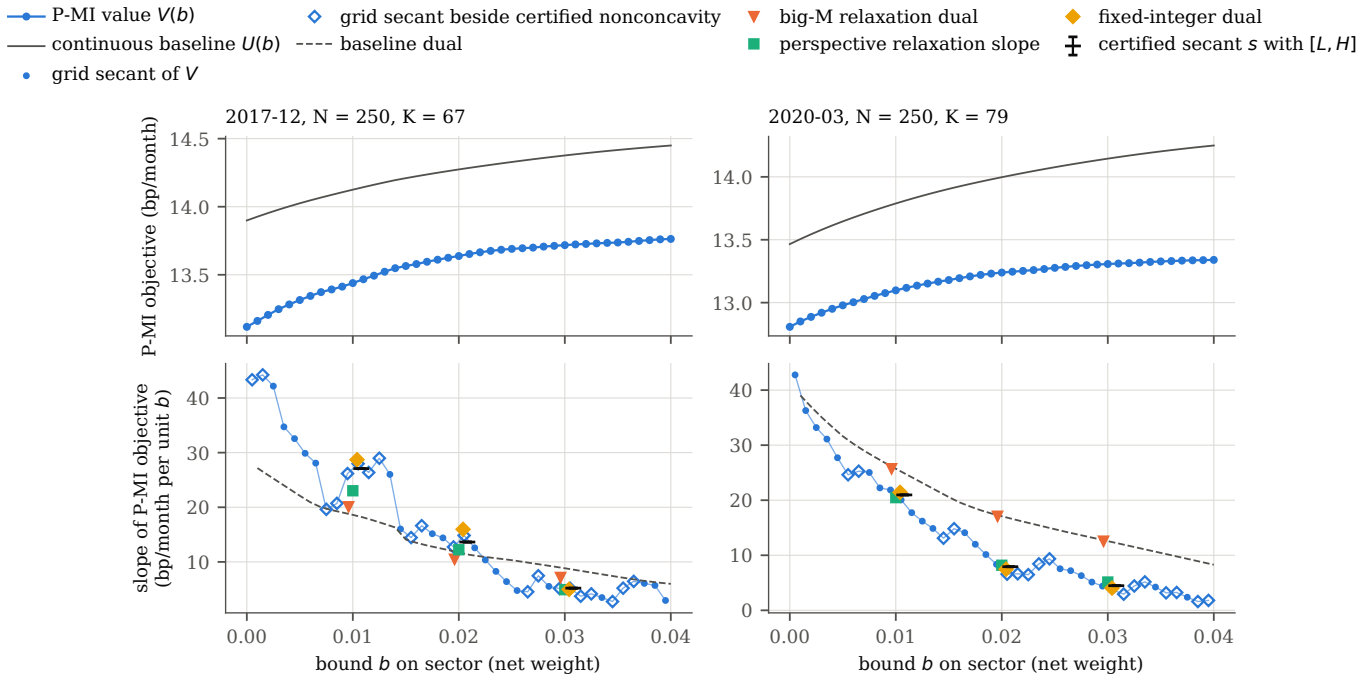


Fig. 4. I2 value functions on the sector bound for two instances (250 names; left 2017-12, right 2020-03). Top: the P-MI value $V(b)$ (combined certification) and the continuous baseline ($\ell = 0$, no cardinality row), as P-MI objective. Bottom: slopes, as P-MI objective per unit of bound: the secant of V between neighbouring grid points (hollow diamonds beside a certified nonconcave grid point), the baseline's dual, and at $0.5b$, b and $1.5b$ the three estimators beside the certified secant s with its interval $[L, H]$ (offset slightly for legibility).

C. Q3: Formulations and Numerics

Setup. N1 solves P-MI at 2017-12 and 2020-03 with $N \in \{100, 250, \text{all eligible}\}$ and $f \in \{0.25, 0.5\}$, at three tightness levels: loose (styles 0.4, beta 0.2), base, and tight (0.025, 0.0125). Each instance is solved by big-M, the perspective at default scaling and the rescaled perspective, at MIPGap 10^{-4} and WorkLimit W^* (144 solves, including a presolve-off repeat at $N = 250$). The criteria are Work and the root gap, (root bound − best known value)/|best known value|, taken after Gurobi's root cut loop. Each cell of Fig. 5 has $n = 12$ instances. Default-scaling results are reported as solver behaviour.

Root gaps and effort. At default presolve the rescaled perspective has the smallest median root gap (4.1×10^{-4}, 3.3×10^{-4}, 1.3×10^{-3} for loose, base, tight) and the least Work (0.66, 0.66, 0.77). Big-M is close in gap (4.7×10^{-4}, 4.4×10^{-4}, 1.1×10^{-3}) with more Work (1.1, 1.0, 1.1). The default-scaling perspective is far behind (0.011, 0.015, 0.022) and does the most Work (1.4, 1.2, 2.1). No solve at default presolve stopped at the WorkLimit. The plain relaxations order as A6 predicts: on every instance the perspective relaxation's value is at most big-M's (0 violations), by a median margin of 0.057 of $|V|$ (range 0.018–0.29). At base, the median plain gaps are 0.059 for big-M and 1.8×10^{-3} for the perspective. Big-M's root gap is two orders of magnitude below its plain relaxation's, and Gurobi's presolve does that work. At $N = 250$ ($n = 12$ per arm), turning presolve off raises big-M's median root gap from 6.0×10^{-4} to 0.057 (its plain gap is 0.060) and

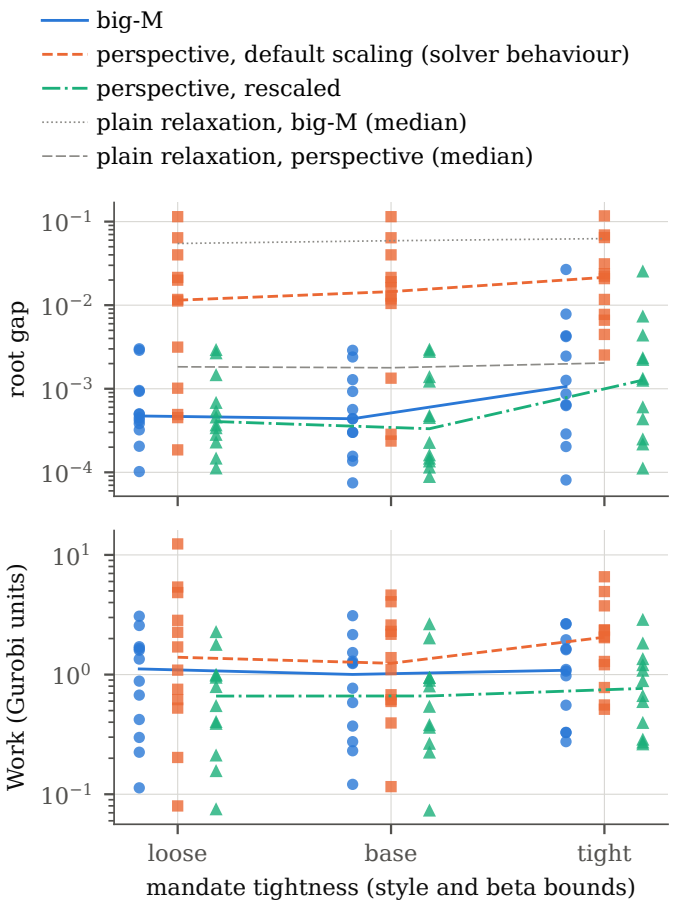


Fig. 5. N1: root gap after the root cut loop (top) and Work (bottom) against mandate tightness, default presolve; one marker per instance ($n = 12$ per cell), medians connected. The default-scaling perspective is solver behaviour. Thin gray lines mark the median plain-relaxation gaps of big-M (dotted) and the perspective (dashed).

its median Work from 1.09 to 255, with 6 solves stopped at the WorkLimit. The rescaled perspective moves only from 4.9×10^{-4} to 5.5×10^{-4} (Work 0.766 to 2.28), near its plain gap of 1.4×10^{-3}.

Agreement and cone tolerance. Every arm's polished value lies within every other arm's B_{eff} (0 failures in 144 pairs). The worst $|\Delta V|$ uses 0.97 (big-M against default scaling) and 0.64 (against the rescaled arm) of the larger certified gap. Reruns under the WorkLimit are identical (3 of 3). The default-scaling arm's incumbents violate their cones within tolerance, with an objective effect of median 4.0×10^{-4} and max 6.4×10^{-3} of $|V|$; for the rescaled arm it is at most 8.0×10^{-9}.

Objective scaling and weight scaling. N2(a) repeats the base instances with $N = 250$ at $s_{\text{obj}} = 1$ and 10^4 (24 solves; Table III). Objective scaling governs the absolute tolerances of branch and bound. At $s_{\text{obj}} = 1$ big-M stops at the WorkLimit in 2 instances, with median certified gap 2.0×10^{-3} and median Work 172; at 10^4 the gap is 9.4×10^{-5} and the Work 1.00. Weight scaling governs the cones and does not replace objective scaling. At $s_{\text{obj}} = 1$, at 2017-12 with $K = 67$, the rescaled perspective reported a closed gap while its B_{eff} lay $9.1 \times 10^{-5}|V|$ below big-M's polished value: a false certificate. At 10^4 no incumbent exceeds another arm's bound (0, against 1 at $s_{\text{obj}} = 1$), and no disagreement exceeds the larger claimed gap (0 of 24 pairs). N2(b) solves P-RB at the dates 2008-10, 2013-06, 2017-12, 2020-03, 2025-04 against Clarabel at tolerance 10^{-10}. Gurobi's barrier on the unscaled QCQP errs by up to 3.1×10^{-5} in A^* and 0.057 in the binding duals, and objective scaling reduces these to 7.2×10^{-7} and 9.5×10^{-3}. Normalizing the risk row to right-hand side one changes nothing for the barrier. The written models differ in exactly 1 row on all 10 date and scaling pairs, and A^* is identical with and without it at default settings (10 of 10) and with Gurobi's scaling off (10), but not with presolve off, where it differs by up to 1.6×10^{-5}. Finally, in 7 of 45 perspective relaxations, 11 names hold both legs (largest smaller leg 1.4×10^{-4}), each with net weight below ℓ and $z^+ + z^-$ above $|w^+ - w^-|/\ell$ (11 of 11, as predicted): the relaxation hedges to cheapen its cone.

V. Discussion

TODO(Isaiah), V, 1.0 page; DESIGN §11.4 and the S4 addendum item 6 (claim → evidence):

- Modeling choice, fixed risk budget against fixed λ: costs at equal risk; mean-variance recovered at λ^* (A1) → 3.8×10^{-8}.
- Modeling choice, λ^* fixed within (date, N) for P-MI: a Lagrangian relaxation of the risk row, exact only without integers; the MIQP's volatility can fall below the budget → Sec. II.
- Modeling choices: single period with $w_0 = 0$; the six-factor, 252-day risk model with the residual floor; the momentum construction and winsorization; the base mandate values; the 6% cap and ℓ in P-MI; costs as assumptions → Table I.
- Limitations: the disclosures below.
- Trade-off, alpha against exposure: the cost curves → Fig. 1, 0.744, 0.961.
- Trade-off, formulation strength against per-node cost: tighter at the root, but cone nodes → 0.057; median nodes at base 36.5 (big-M), 6.5 (rescaled), 270.5 (default).
- Trade-off, tightness against effort → Fig. 5.
- Trade-off, tolerance against runtime: MIPGap 10^{-6} (I1) against 10^{-4} (N1); Work at $s_{\text{obj}} = 1$ → 172 against 1.00.
- What each estimator measures (A8), and why the errors depend on δ → 0.052 against 0.016 at 2017-12.
- Two failure modes: a support change within the step for the fixed-integer dual (P3 0.056; P1 6.7×10^{-12}), and relaxation looseness for the relaxation slopes (Spearman 0.64, 0.50; perspective per class 0.011, 0.014) → results/s3b_analysis/ item 3.
- The I2 near-tie (Decision 44): median secant 1.364×10^{-3} (sector) against 1.316×10^{-3} (momentum); un-

TABLE III

N2 numerics. Top, N2(b): P-RB at the dates 2008-10, 2013-06, 2017-12, 2020-03, 2025-04, by solver, risk-row normalization and objective scaling, worst over the dates against Clarabel at 10^{-10}. Bottom, N2(a): P-MI at the base instances with $N = 250$, by arm and s_{obj}, at MIPGap 10^{-4}; certified gaps are B_{eff}-based.

convex solver (normalized, scaled)	$\lvert\Delta A^*\rvert/A^*$	$\lVert\Delta w\rVert_1/\lVert w\rVert_1$	dual err.	solved
Clarabel (no, no)	1.7×10^{-7}	2.6×10^{-5}	2.2×10^{-4}	5/5
Clarabel (no, yes)	1.7×10^{-8}	1.3×10^{-5}	2.2×10^{-4}	5/5
Clarabel (yes, no)	2.7×10^{-7}	7.4×10^{-5}	2.5×10^{-4}	5/5
Clarabel (yes, yes)	4.3×10^{-9}	1.1×10^{-5}	2.9×10^{-5}	5/5
Gurobi barrier (no, no)	3.1×10^{-5}	1.9×10^{-3}	5.7×10^{-2}	5/5
Gurobi barrier (no, yes)	7.2×10^{-7}	1.2×10^{-3}	9.5×10^{-3}	5/5
Gurobi barrier (yes, no)	3.1×10^{-5}	1.9×10^{-3}	5.7×10^{-2}	5/5
Gurobi barrier (yes, yes)	7.2×10^{-7}	1.2×10^{-3}	9.5×10^{-3}	5/5
SCS (no, no)	5.1×10^{-4}	4.1×10^{-4}	2.7×10^{-3}	5/5
SCS (no, yes)	2.7×10^{-6}	2.9×10^{-5}	1.0×10^{-4}	5/5
SCS (yes, no)	3.6×10^{-6}	5.6×10^{-4}	2.4×10^{-3}	5/5
SCS (yes, yes)	2.6×10^{-6}	1.5×10^{-5}	9.1×10^{-5}	5/5
MIQP arm, s_{obj}	median $(B_{\text{eff}} - V)/\lvert V\rvert$[a]	max cone effect	V above other B_{eff}	B_{eff} below other V
big-M, 1	2.0×10^{-3}	–	1	0
big-M, 10000	9.4×10^{-5}	–	0	0
perspective (default scaling), 1	1.2×10^{-3}	4.9×10^{-3}	0	0
perspective (default scaling), 10000	1.2×10^{-3}	4.8×10^{-3}	0	0
perspective (rescaled), 1	1.0×10^{-6}	0	0	1
perspective (rescaled), 10000	9.0×10^{-5}	8.2×10^{-18}	0	0

[a] $B_{\text{eff}} = \max(B, V) + 10^{-6}\lvert V\rvert$-based, so floored at 10^{-6}.

der the perspective slope the order flips, 1.222×10^{-3} against 1.306×10^{-3}.

- Unexpected: presolve does big-M's root work → 6.0×10^{-4} against 0.057; Work 1.09 against 255.
- Unexpected: the false certificate at $s_{\text{obj}} = 1$ → 9.1×10^{-5}.
- Unexpected: the risk-row normalization is absorbed by presolve (supported, not proven) → 10 of 10, 1.6×10^{-5}.
- Unexpected: the hedged legs; per-leg cones as an extension → 11, 11.
- K29b-ext: slope accuracy holds on every I1 case, not only the gated ones → 36 of 36.
- The earlier iteration's lessons in one or two sentences, without numbers: the fixed-risk formulation and the scaling policy.
- Extensions: platform coordination by exposure prices (decomposition); multi-period rebalancing (MPC); robust alpha (SOCP); a Lagrangian bound for the cardinality row.

VI. Conclusion

TODO(Isaiah), VI, 0.25 page (claim → evidence):

- Q1: neutrality is cheap for a weakly aligned alpha; caps and sector bounds carry the cost → 0.16, 0.964.
- Q2: duals stop being prices once integers enter; the fixed-integer dual is exact only on a stable support → 6.7×10^{-12}, 0.056, Table II.
- Q3: the perspective formulation helps only with its weights rescaled, and objective scaling is still needed → 3.3×10^{-4}, 0.015, 9.1×10^{-5}.
- Extensions → Sec. V.

Disclosures and AI-Use Acknowledgment

TODO(Isaiah), disclosures, DESIGN §11.5 (claim → evidence):

- Universe: current S&P 500 constituents (snapshot 2026-10-16), so survivorship and look-ahead in the universe; GICS sectors are not point-in-time → results/data_report.md.
- Yahoo adjusted data; the JCI and DHR errors masked; a day is valid only with a return and dollar volume of at least $1M, a stock eligible with 80% valid days in its window; 1 multi-session return enters the regressions as a one-day return → results/data_report.md.
- One-pass winsorization leaves $\lvert z\rvert$ above 3: max 5.47 (2024-04), at 240 of 240 month-ends, 1936 of 107592 name-dates → results/disclosures/.
- Single period, $w_0 = 0$; borrow and trading costs are assumptions → Table I.
- No backtest and no performance claims; the alpha is a simple input signal.
- The parameter-setting rules of DESIGN §8 (the design pilot: s_{obj}, W^*, the K grids, the I1 constraints) → experiments/configs/integer.yaml.
- The earlier iteration, without numbers.

TODO(Isaiah), AI-use acknowledgment, from docs/AI_USE.md's v2 log (claim → evidence):

- Claude (chat) designed v2 and extracted the repository; proposed most design decisions, check text, DESIGN edits and gate notes; sketched derivations; drafted the proposal text. Isaiah chose among the proposals, edited and signed every gate note, and ran the commands.

- Claude Code, with the result-checker and python-reviewer subagents, wrote and ran the code, tests and checks, built the figures and tables, and drafted Sections II–IV of this report (Mode A); Isaiah revises them and checks the derivations by hand.

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